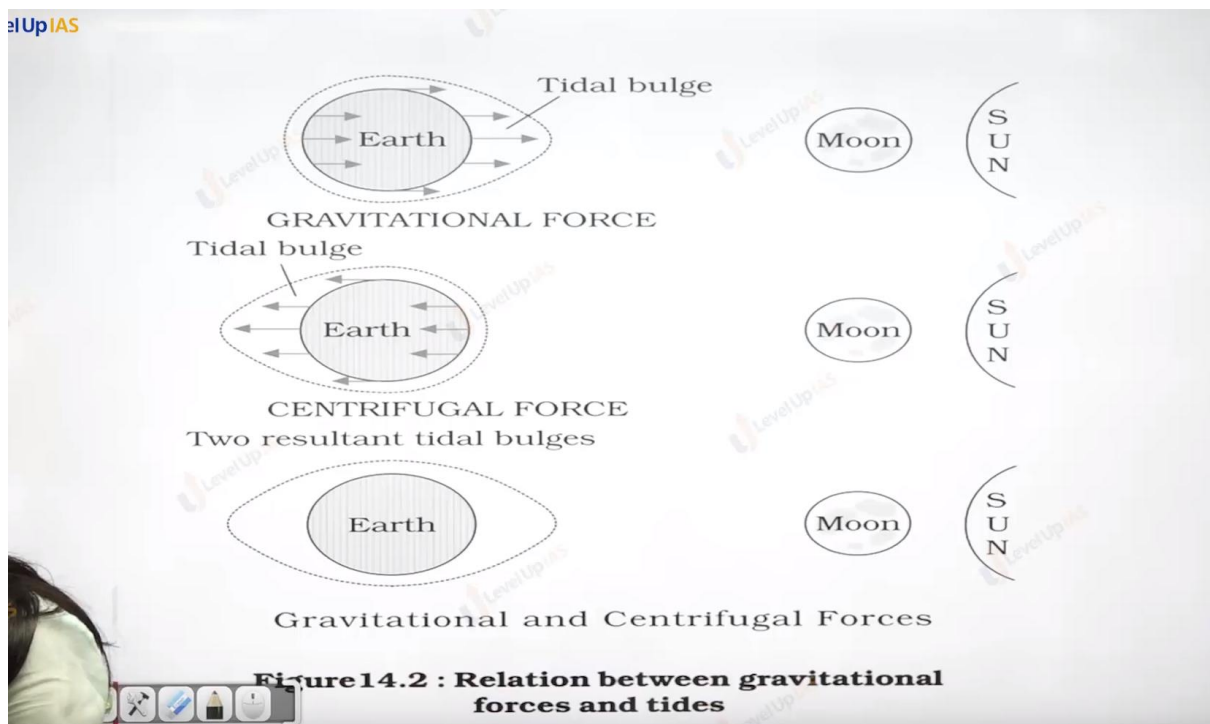
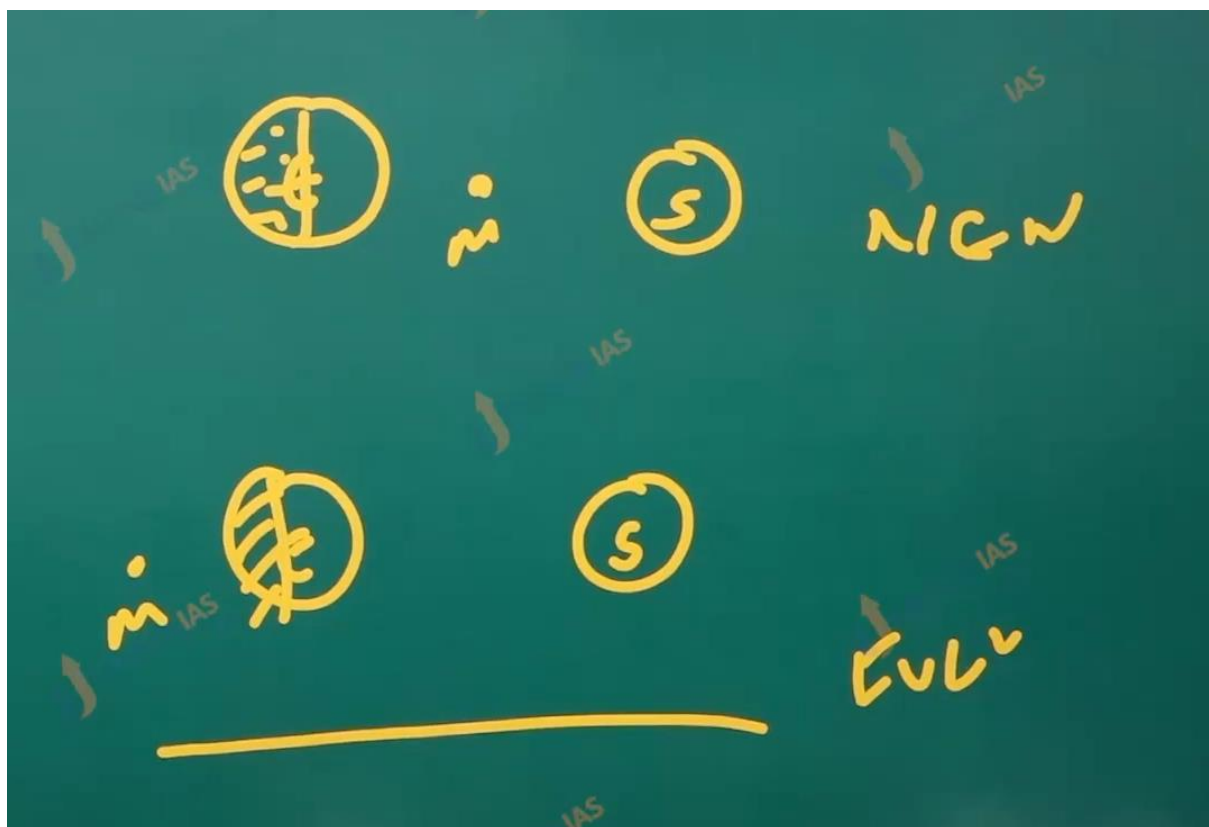
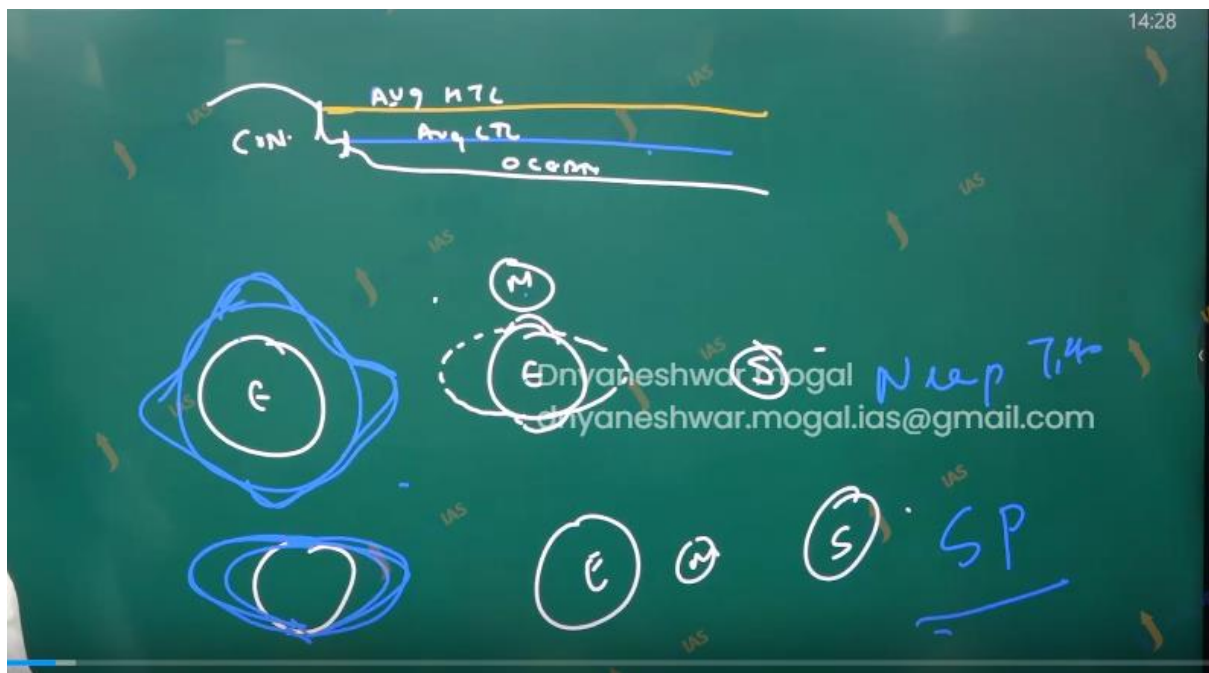


(27) Waves, Tides

Tides:

1. Tides refers to vertical movement of water nearly each place on the earth experiences two high tide tow low tide
2. Reasons for formation of tides are gravitational pull of sun, gravitational pull of moon and the centrifugal force of earth as counter force to gravitational pull of sun and moon.
3. The gravitational pull of moon plays an important role because of its nearness to earth.





Type of tide

1. Basis of sun and moon:

- a. The position of sun and moon are in straight line. At this position the height of high tides is above the average high tide level. And
 - b. The low tides are quite below the average low tide level. Therefore, the difference between high tide and low tide is very high.
 - c. They occur 2 in the month once full moon day and new moon day.
-

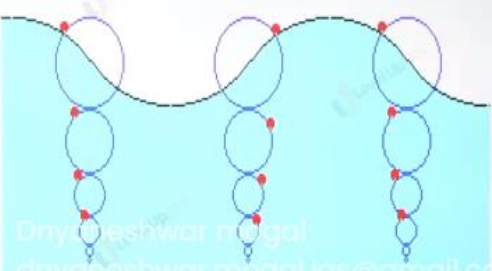
Neap type:

-
1. Normally there is 7 days interval between spring and neap type.
 2. Here in the neap type the sun, earth and moon are at a right angle to each other.
 3. The forces of sun and moon tend to counter acts one another the moon attractions Tho more than twice as strong as that of suns is diminished by the counter acting force of sun gravitational pull.
 - 4.
-

Importance of tide.

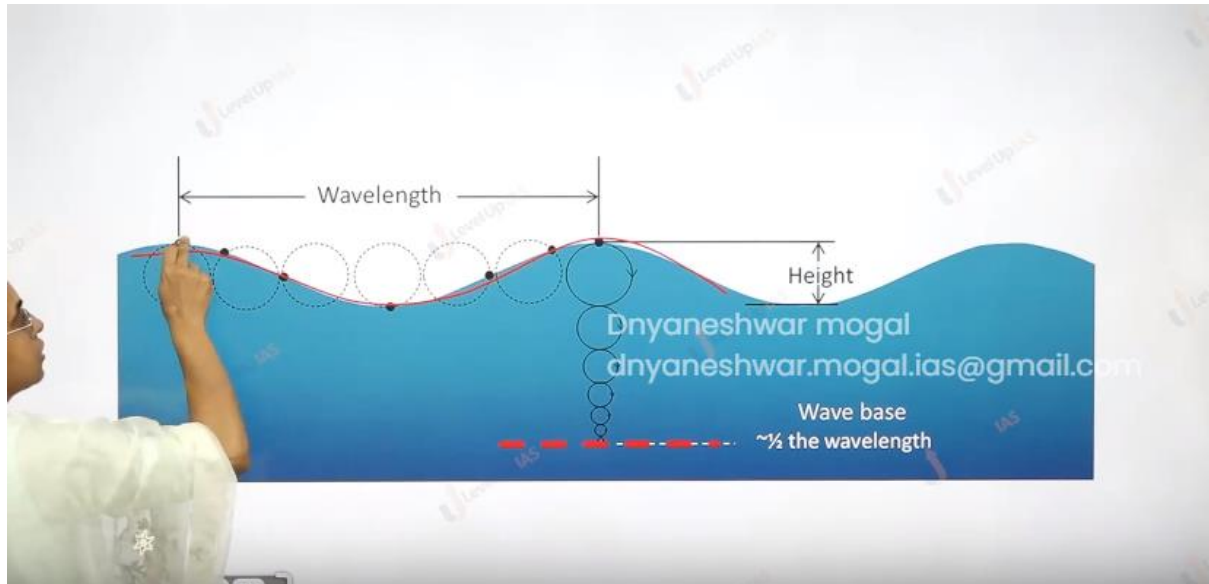
-
1. It helps navigator and fisher man.
 2. Tidal energy can be utilized to supplement the present energy mixed.
 3. Tidal plants in the gulf of combat, gulf of Kutch, Andaman Nicobar Island, Durgaduani in Sundarbans.
 4. Tides also help in desiltation along the coast.
 - 5.
-

Waves

Waves:

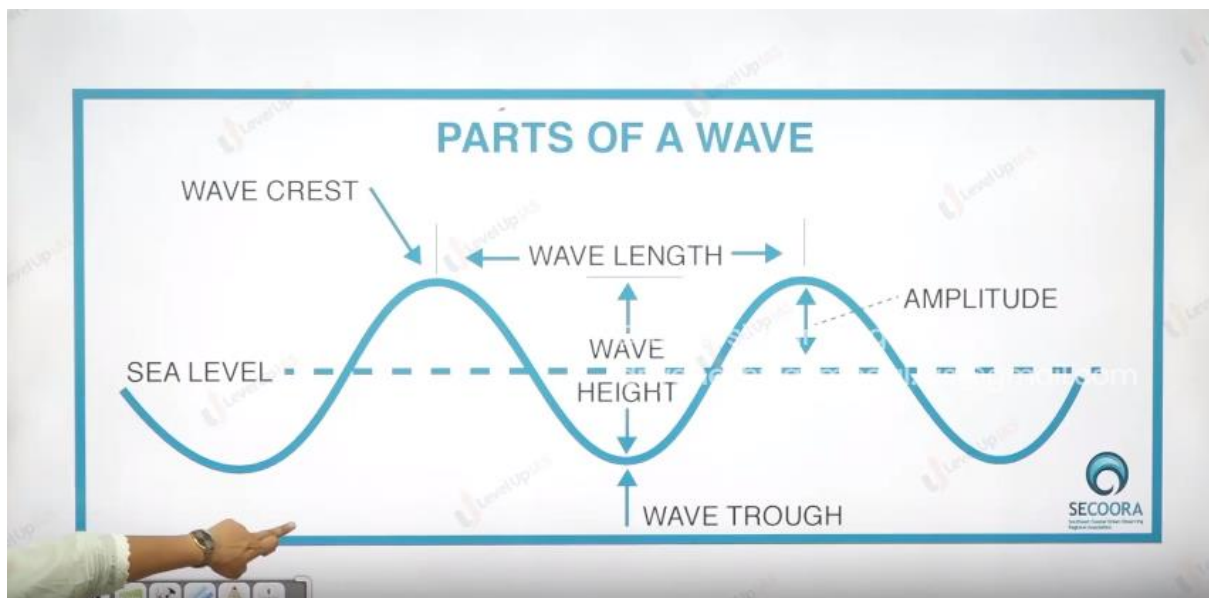
1. Waves are disturbances or ripples formed overseas surface as energy moves from one place to another.
2. Do not involve movement of water particles it is the only the energy which moves.
3. Formed when producing force remains smaller than the resisting force due to which the energy is unable to carry the water particle and and 's only successful in creating disturbances



Waves:

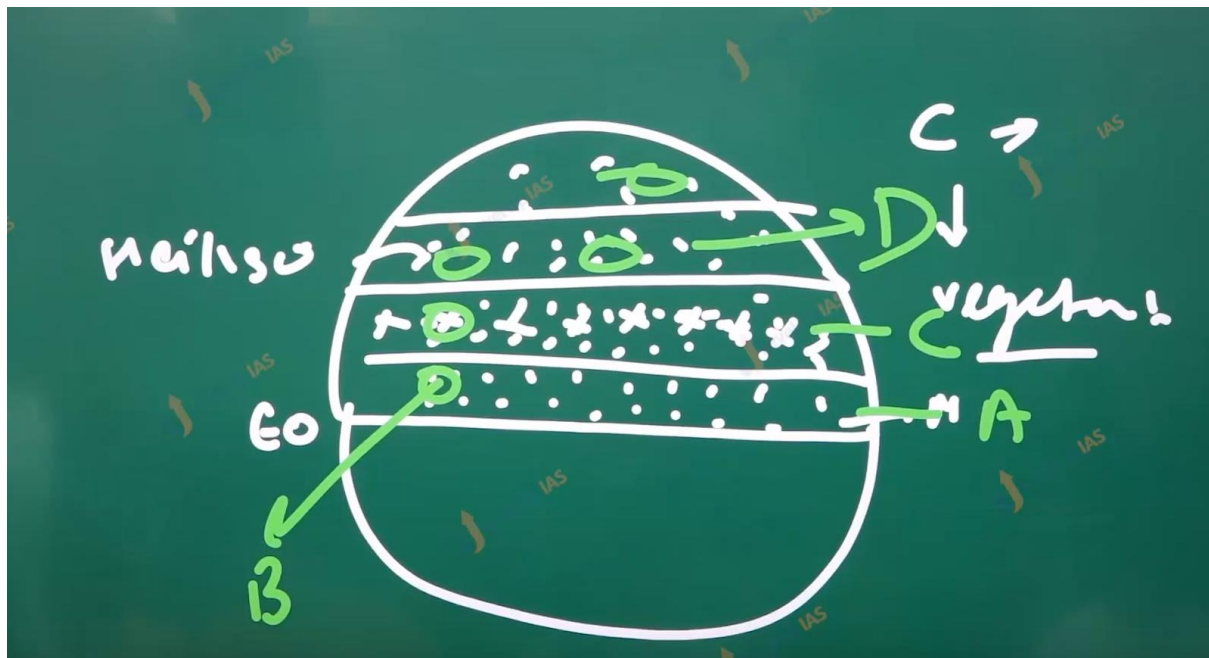
1. Waves are disturbances or ripples on the ocean surface as energy moves from one place to another.

2. It formed when the energy producing source is smaller than the resisting force due to which energy is unable to carry the water particles and it is successful in only creating the disturbance.
 3. In waves it is energy that moves not the water as such which moves across ocean surface.
 4. Water in the wave moves in the circular manner / small circular as the wave passes. As long as wave's base is half of wavelength the wave maintains its shape but as soon as coast appears the wave starts breaking.
 - 5.
-

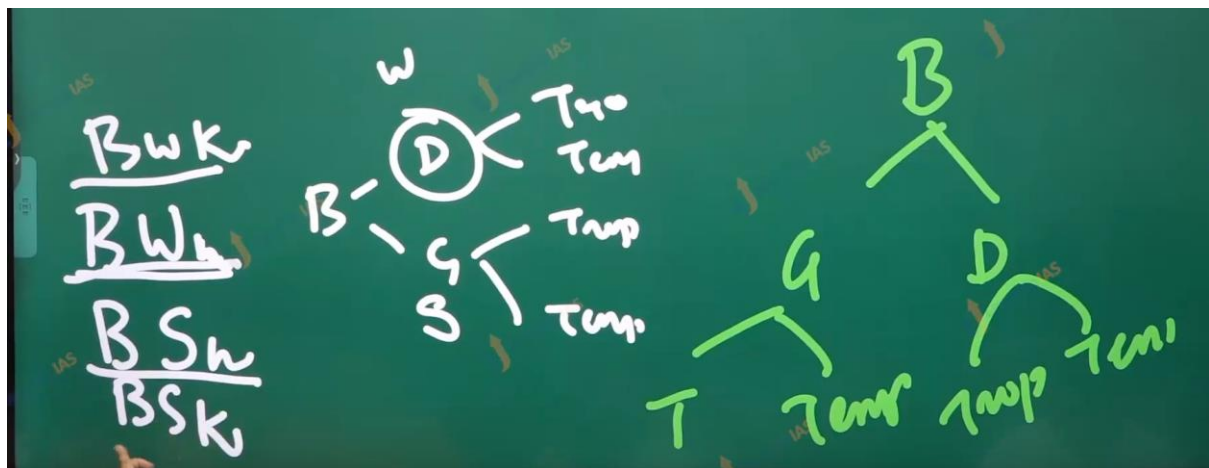


Parts of waves:

-
1. The wave can maintain its shape as long as the depth is half of wavelength or greater than half the wavelength.
 2. On approaching the coast, the wave breaks as depth of ocean floor becomes less than the half of wavelength of wave.
-



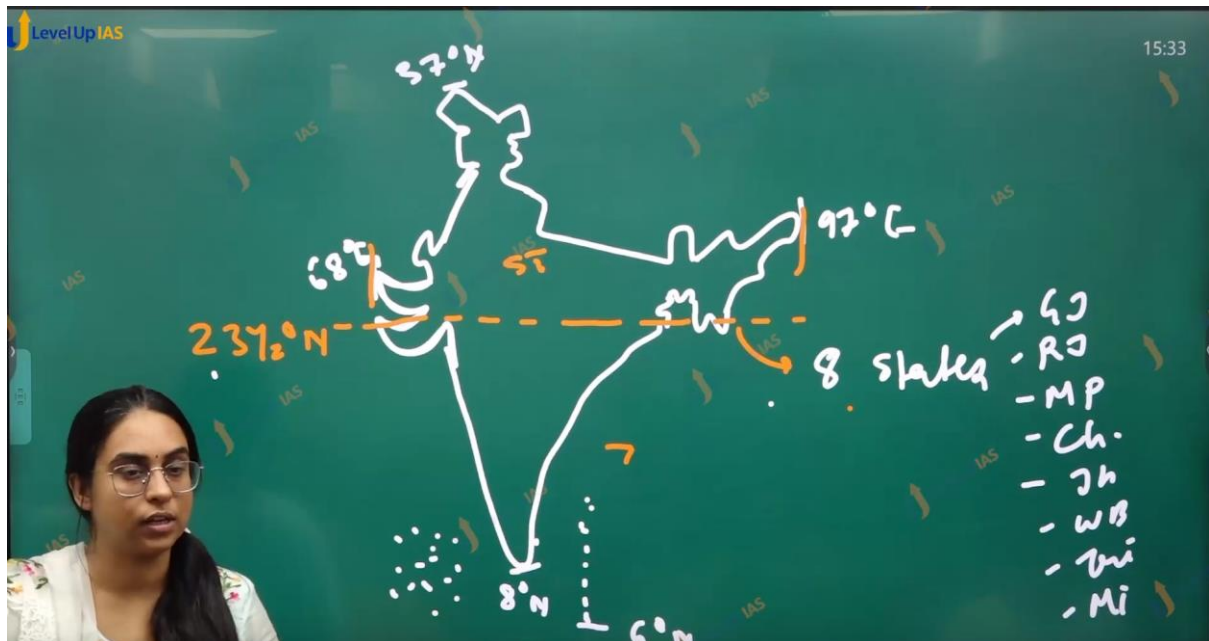
World and climate and climate change:

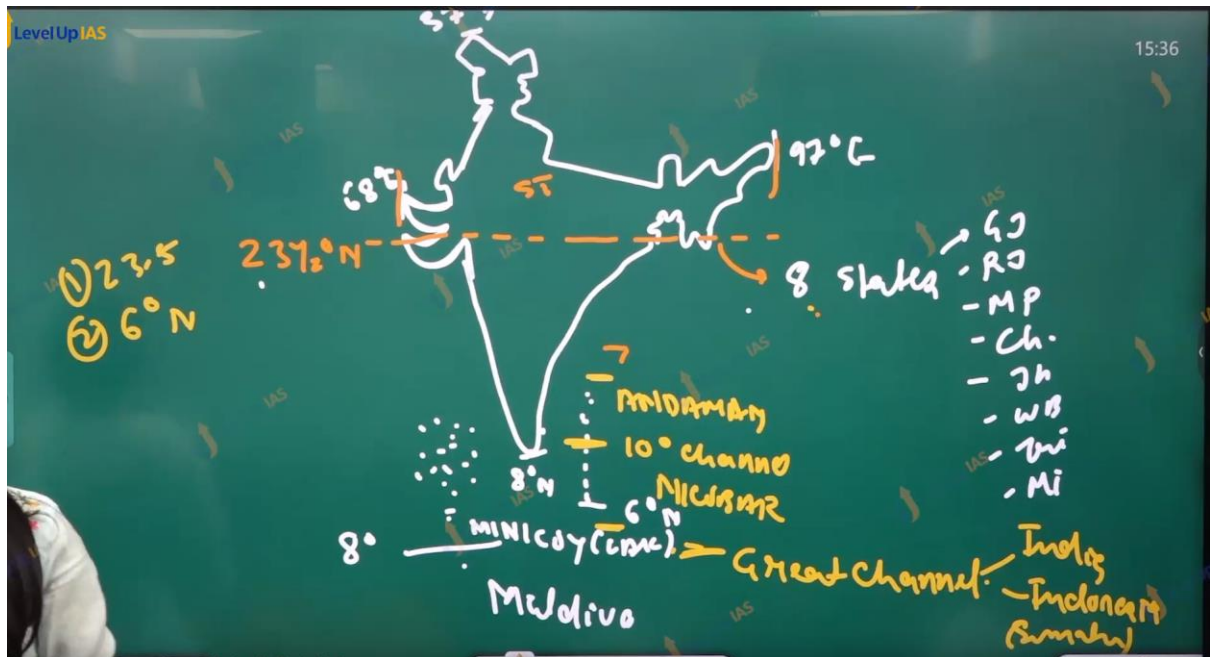
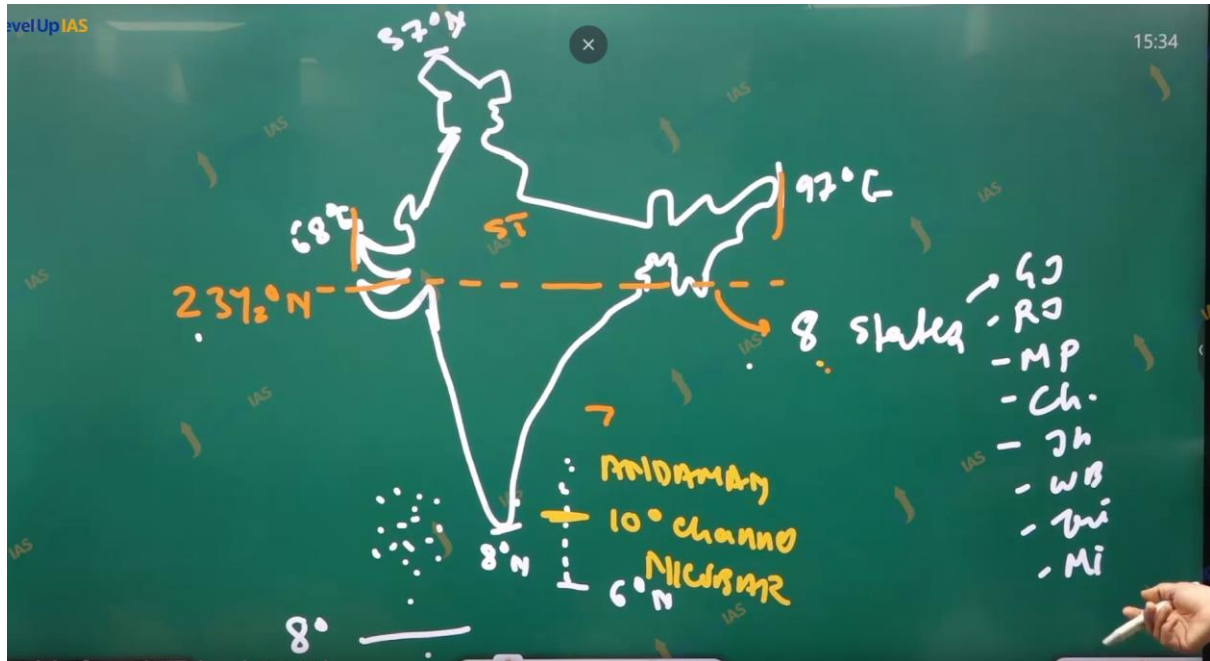


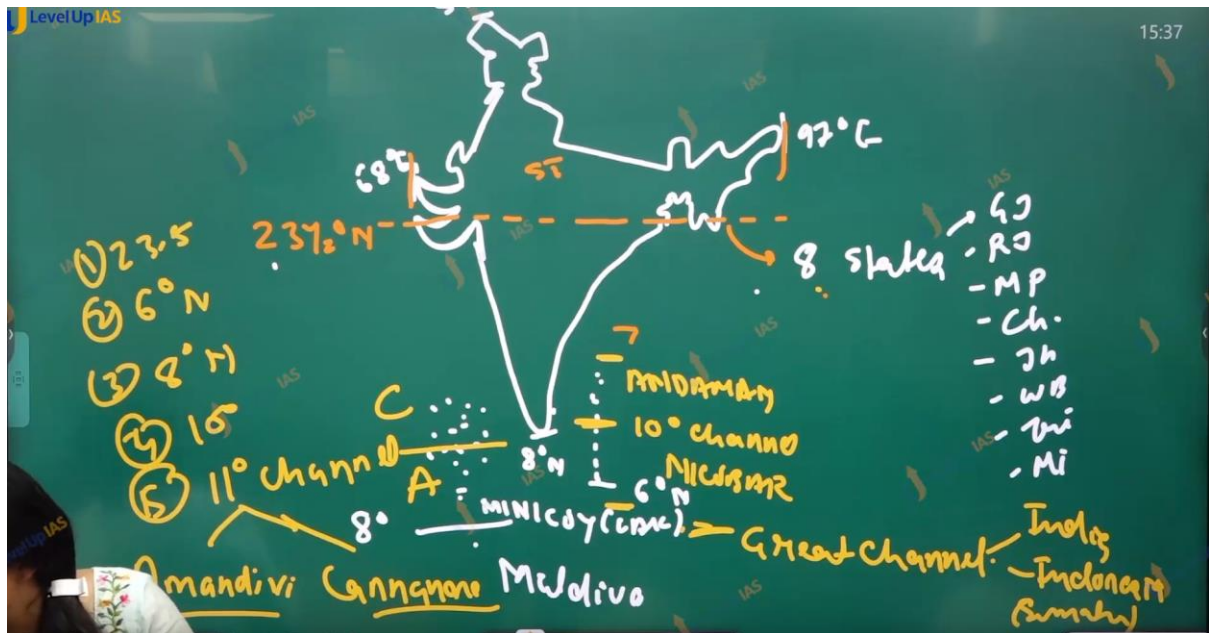
Later given to the weather pattern in the world.

Important Lines

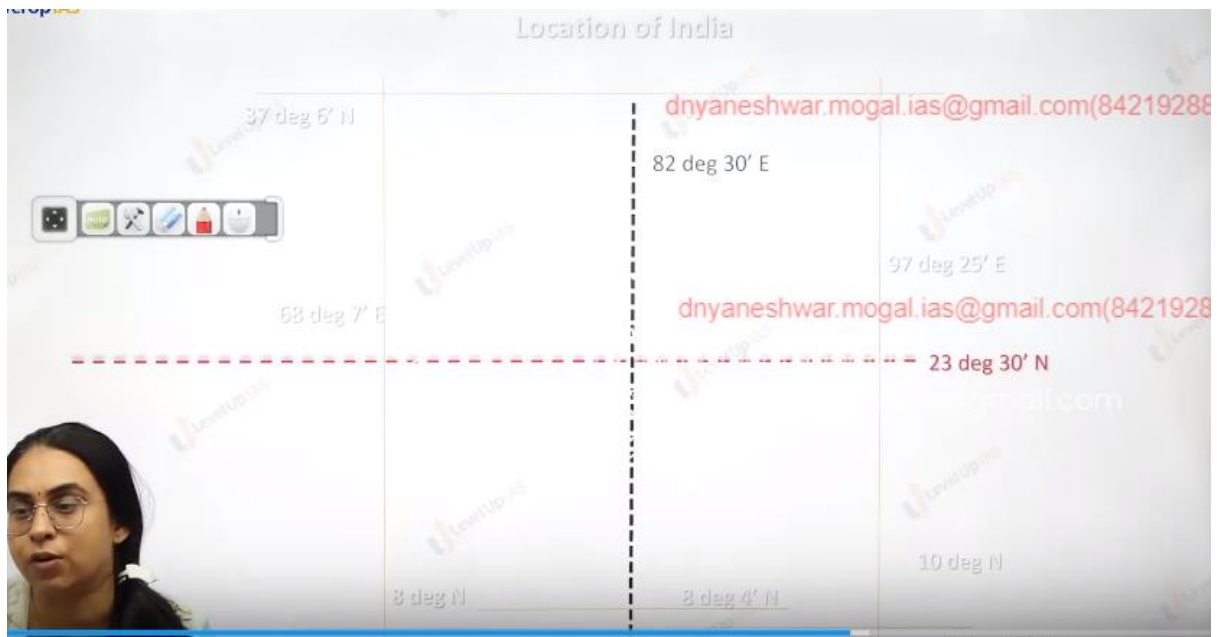
- ☐ Tropic of cancer passes through which states?
- IST passes through which states?
- North, South, West, East point of India







Amindivi and Cannanore.



- Recognize the Northernmost point, Southernmost point, Westernmost point, Easternmost point.
- Recognize IST



Neighbors of India

Facts about India's geography

1. India is 7th largest country in the world. India has 2.4% of the world's land and 17% of the world's population.
 2. 17% of the world's livestock. And 4 % water resources.
 3. IST follows 82 deg 30' E it passes through 23 deg 30'N
 4. Neighbours are Bangladesh, China, Pakistan, Nepal, Myanmar, Bhutan, Afghanistan.
 5. Country size wise. Russia, Canada, China, Brazil, Australia and India.
-

2014

Which one of the following pairs of islands is separated from each other by the "Ten Degree Channel"?

- (a) Andaman and Nicobar
- (b) Nicobar and Sumatra
- (c) Maldives and Lakshadweep
- (d) Sumatra and Java

Answer is a

2015



Which one of the following pairs of States of India indicates the easternmost and westernmost State?

- (a) Assam and Rajasthan
- (b) Arunachal Pradesh and Rajasthan
- (c) Assam and Gujarat
- (d) Arunachal Pradesh and Gujarat



2015



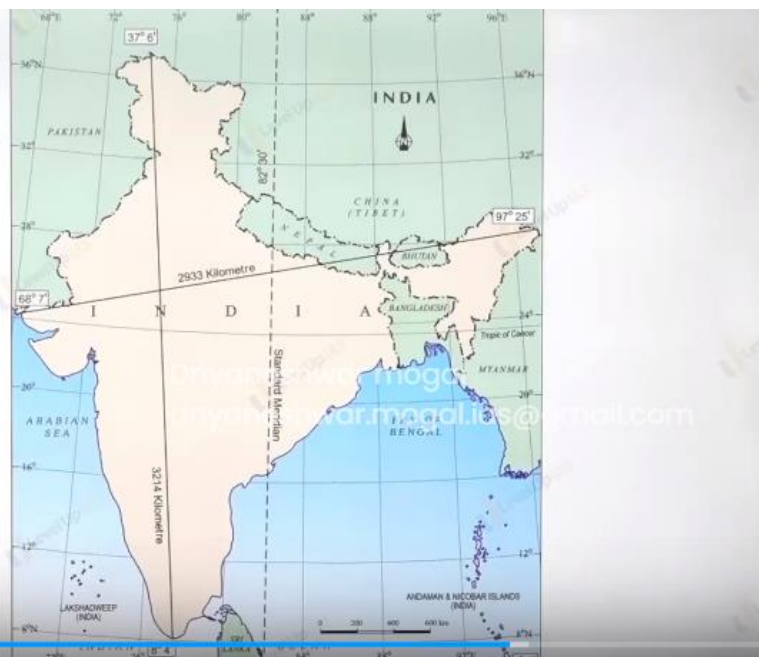
Which one of the following pairs of States of India indicates the easternmost and westernmost State?

- (a) Assam and Rajasthan
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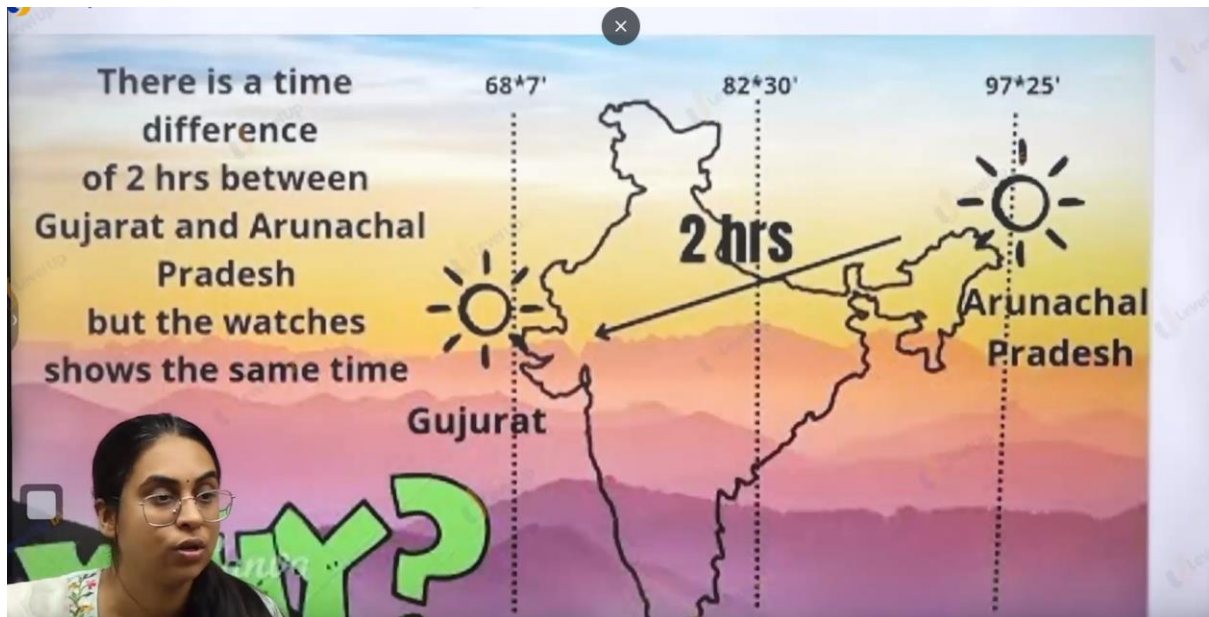
SOME BASIC INFORMATION

- **Latitudinal and longitudinal extend:** 30 degree North and South and 30 degree East and West
- **Distance N-S:** 3214 km
- **Distance: E-W:** 2933 km
- Reason: Distance between two longitudes decrease towards the poles whereas distance between two latitudes remains the same



Some basic information's

1. Latitudinal and longitudinal extents is 30 degrees. The north south extents are more 3214 where east western 2933 this is because the distance between two longitudes is degree towards the pole whereas distance between the two latitudes
- 2.



LevelUp IAS

Need of two time Zone:

- **India is geographically large:** Need to have multiple time-zones
- **Northeast lose important daylight** which can be used productively as the sun rises as early as 4 am in summer and offices open at 10.
- **People's productivity** and efficiency follows a biological clock that is synchronized with the daily light-dark cycles.
- **Wastage of electricity:** Government offices: 9 am or 10 am so there is wastage of electricity
- **Northeast have complained** about single time zone on their lives. North Eastern States have proposed two time zone

There is a time difference of 2 hrs between Gujarat and Arunachal Pradesh but the watches shows the same time

68°7' 82°30' 97°25'

2 hrs

Gujarat Arunachal Pradesh

WHY?

LevelUp IAS

Benefits of two time Zone

- Lead to greater efficiencies among the workforce and on energy consumption.
- Advancing IST by just a half hour would result in saving 2.7 billion units of electricity every year by using the waste daylight hours.
- Reduction in energy consumption will significantly cut down India's carbon footprint boosting India's resolve to fight climate change.
- People will be able to work better and plan better, according to natural cycles.
- Two time zones will allow aligning standard time with daylight time.

WHITHER TIME ZONES

India would save **2.7 billion units of electricity** every year by shifting the IST meridian eastward (from 82.5E longitude in Uttar Pradesh to 90E near the Assam-Bengal border).

India stands to gain little, and could in fact suffer from negative energy saving, if two time zones are introduced

Advancing IST by 30 minutes saves more electricity than the corresponding proposals for introducing daylight saving time or introducing two time zones in the country staggered by one hour,"

% of total energy consumption that can be saved if time is advanced by 30 minutes

Karnataka	0.49
Tamil Nadu	0.21
Andhra Pradesh	0.28
West Bengal	0.64
Madhya Pradesh	0.68

4 state that share the border with Myanmar Arunachal Pradesh, Nagaland Manipur and Mizoram.

Find out the state that share border with Pakistan China and Nepal. Bangladesh Bhutan.

Pakistan: Jammu and Kashmir, Rajasthan, Panjab, Gujrat.

Nepal: Uttarakhand, Utter Pradesh, Bihar, West Bengal, Sikkim.

China: Ladakh, Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh.

Bangladesh: Assam, West Bengal, Mizoram, Meghalaya, and Tripura.

Bhutan: Assam, Arunachal Pradesh, Sikkim, and West Bengal.

